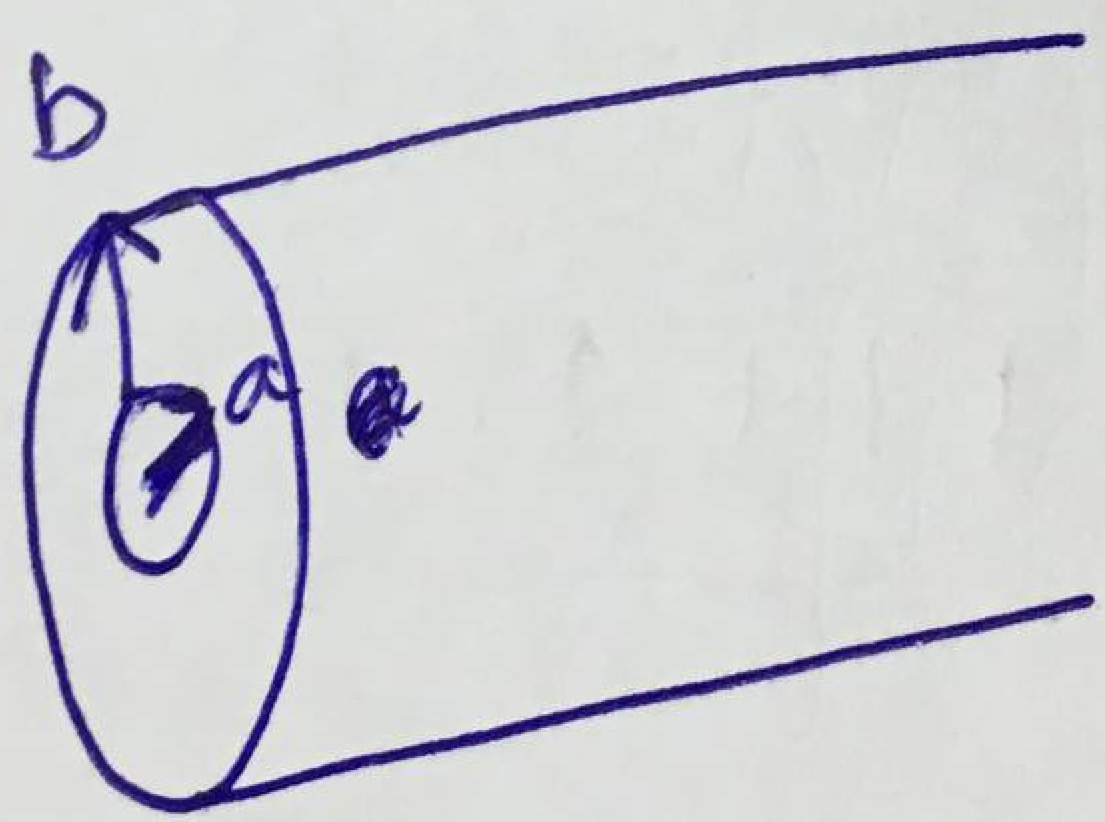
Q1(a) A cylindrical-capacitor which has inner and outer radii of $a = 5mm$ and $b = 15mm$, respectively. If $V(a) = 100\ V$ and $V(b) = 0\ V$ (GND), calculate i) V, ii) $\mathbf{E}$, and iii) $\mathbf{D}$ at $\rho = 10\ mm$ and iv) ρ_s on each plate. Take $\varepsilon = 2$ and draw the diagram as well.

(b) Determine the capacitance if length of cylindrical capacitor is $2cm$.

CLO 05, PLO 01 → SLO 1.1

$$= V_0 \frac{\ln(b/\rho)}{\ln(b/a)}$$



$$100 \times \frac{\ln(3/2)}{\ln(3)}$$

$$\varepsilon = \varepsilon_r \varepsilon_0$$

Q2 A circular loop located on $x^2 + y^2 = 9,\ z = 0$ carries a direct current of $10\ A$ along $\boldsymbol{a}_\varphi$. [10]
Determine $\boldsymbol{H}$ at $P(0,0,4)$.

CLO 06

$I = 10A$

$r = 3$

$$H = \frac{I}{2\pi R} a_\phi$$

$$H = \frac{10}{2 \times \pi \times 3} a_\phi$$

$$\oint \frac{60 \cancel{\pi} \vec{a_r}}{2 \cdot \cancel{\pi} \times 3}$$

$$= \oint 10 \vec{a_r}$$

Name:

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Q3 If $\boldsymbol{H} = y\boldsymbol{a}_x - x\boldsymbol{a}_y \ A/m$ on plane $z = 0$;

(a) Determine the current density and [5]

(b) Verify Ampere's law by taking the circulation of **H** around the edge of the rectangle $z = 0, 0 < x < 3, -1 < y < 4$. [10]

CLO 07

$H = ya_x - xa_y \ A/m$

$$J = \nabla \times H = \begin{vmatrix} a_x & a_y & a_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} a_x & a_y & a_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & -x & 0 \end{vmatrix}$$

$\partial(y))) + \vec{a}_z($

Name:

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Q4 A charged particle moves with a uniform velocity of $4\boldsymbol{a}_x\ m/s$ in a region where $\mathbf{E} = 20\boldsymbol{a}_y\ V/m$ and $\boldsymbol{B} = \mathrm{B}_o\boldsymbol{a}_z\ Wb/m^2$. Determine B_o such that the velocity of the particle remains constant (i.e acceleration is zero). [1

CLO 08

$v = 4a_x\ m/s$

$E = 20a_y\ V/m$

$B = B_o a_z\ Wb/m^2$

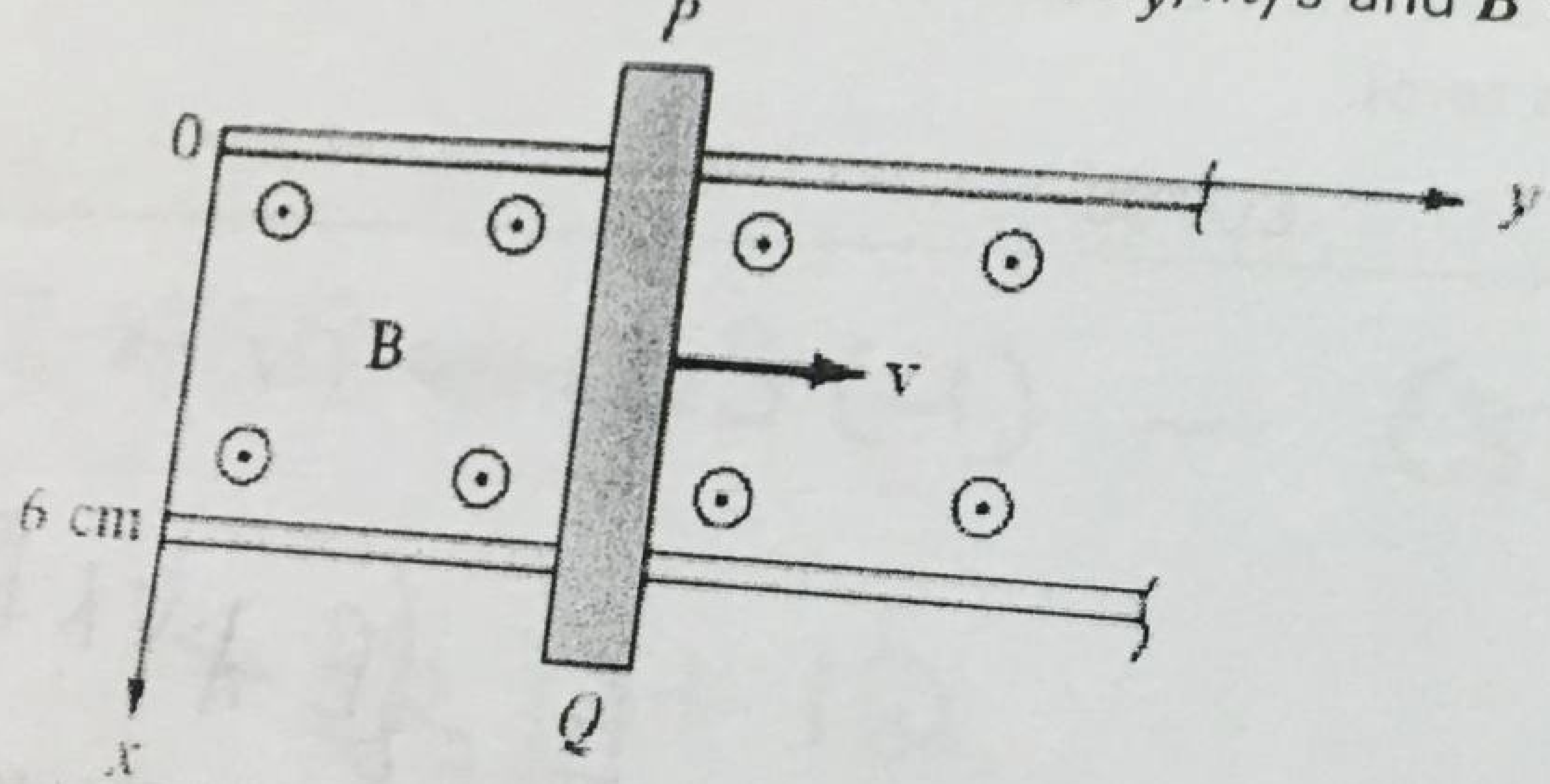
$D \to \rho_v \to J$

$\nabla \times E = \begin{vmatrix} a_x & \cancel{a_y} & \cancel{a_z} \\ & & \partial/\partial z \end{vmatrix}$

Q5 A conducting bar can slide freely over two conducting rails as shown in the figure. Calculate the induced voltage in the bar:

(a) If the bar is stationed at $y = 8\ cm$ and $\mathbf{B} = 4\cos 10^6 t\ \mathbf{a}_z\ mWb/m^2$ [5]

(b) If the bar slides at a velocity $\mathbf{v} = 20\mathbf{a}_y, m/s$ and $\mathbf{B} = 4\mathbf{a}_z\ mWb/m^2$ [5]



CLO 09

b) $V = 20a_y\ m/s$

$B = 4a_z\ mWb/m^2$

$emf = \oint (\vec{V} \times \vec{B}) \cdot dL$

$\vec{B} = | a_x \quad a_y \quad a_z |$

a) $B = 4\cos 10^6 t\ mWb/m^2$

$y = 8 \times 10^{-2}\ m$

Name:

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Roll. No:

Q7 Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\varepsilon_{r1} = 4$ and for $z \leq 0$, $\varepsilon_{r2} = 3$. A uniform electric field $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z$ kV/m exists for $z \geq 0$.

(a) Find $\mathbf{E}_2$ for $z \leq 0$. [4]

(b) Find the energy densities in $[J/m^3]$ in both dielectrics. [6]

CLO 04

$E_1 = 5a_x - 2a_y + 3a_z$

$\varepsilon_0 E$

Q6 In a medium characterized by $\sigma = 0, \mu = \mu_0, \varepsilon = \varepsilon_0$ and $\boldsymbol{E} = 20\sin(10^8 t - j\beta z)\,\boldsymbol{a}_y$ V/m, calculate β and $\boldsymbol{H}$. ? [10

CLO 09

$D = \varepsilon E$

$\Rightarrow D = \dfrac{20\sin(10^8 t - j\beta z)}{8.85\times 10^{-12}}\, a_y$ V/m

$\nabla \times H = J + \dfrac{\partial D}{\partial t}$

$\nabla \times H$ ...

You don't have H.